A DATA VISUALIZATION PROJECT

FOOD ORDERING BEHAVIOUR AND CONSUMER TRENDS

Tableau Dashboard, Data Story and Flask Web Integration

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Academic Year	2026-27

Skill Wallet Project Documentation

1. Introduction

Food ordering platforms generate large amounts of customer and order-level data. Analysing this data can reveal patterns in customer preferences, meal choices, city-wise demand, ratings, restaurant segments and order value. However, raw records are difficult to interpret directly.

This project, Food Ordering Behaviour and Consumer Trends, analyses a food-ordering dataset using Tableau. The data has been transformed into multiple visualizations, combined into an interactive dashboard, and organized into a Tableau Story. The published Tableau views are also embedded into a Flask web application so that the analysis can be accessed through a dedicated website.

2. Problem Statement

Food-ordering data contains multiple attributes such as age, city, cuisine, meal type, company type, restaurant type, customer rating, order value and delivery time. When these records are viewed only as rows of data, it becomes difficult to compare categories and identify important consumer trends.

There is therefore a need for a clear, interactive and visual solution that can show how customers order food, which cuisines and meal types are preferred, how orders differ across cities and company types, and how ratings, revenue and delivery time are distributed.

3. Objectives

- To analyse consumer food-ordering behaviour across age groups, cities, cuisines and meal types.
- To compare order activity across company type and meal type.
- To analyse customer ratings and revenue contribution by meal type.
- To compare city-wise order value and customer preferences across cuisines.
- To analyse delivery-time distribution across orders.
- To design multiple Tableau visualizations and combine them into one interactive dashboard.
- To organize the major findings into an eight-point Tableau Story.
- To publish the Tableau work and integrate the Dashboard and Story into a Flask web application.

4. Methodology / Project Workflow

The project follows a sequential workflow from the raw CSV dataset to the final Tableau Public views and Flask web application. Each step feeds into the next.

STEP 1	**CSV Dataset**
STEP 2	**Import into Tableau**
STEP 3	**Data Preparation**
STEP 4	**9 Visualizations**
STEP 5	**Dashboard**
STEP 6	**8-Point Story**
STEP 7	**Tableau Public**
STEP 8	**Flask Web Application**

Step 1 - CSV Dataset. The project begins with the food-ordering CSV dataset containing customer, order, cuisine, meal, restaurant, rating, order-value and delivery-related attributes.

Step 2 - Import into Tableau. The CSV file is connected to Tableau and its fields are used as dimensions and measures.

Step 3 - Data Preparation. Required fields, data types, aggregations, calculated fields and filters are prepared for analysis.

Step 4 - Nine Visualizations. Nine main visualization areas are created to represent different consumer and order patterns.

Step 5 - Dashboard. The worksheets are arranged into one interactive dashboard.

Step 6 - Eight-Point Story. Key findings are organized into a Tableau Story.

Step 7 - Tableau Public. The workbook is published so the dashboard and story are available online.

Step 8 - Flask Web Application. The published views are embedded into a Flask website using HTML, CSS and JavaScript.

5. Dataset Description

Item	Details
Dataset name	food_ordering_behaviour_dataset.csv
Number of rows	50,000
Number of columns	19
Main data	Food ordering behaviour and consumer/order attributes
Main measures	Order Value, Delivery Fee, Rating Given, Time Taken To Order
Main analysis	Orders, revenue, ratings, cuisines, meal types, cities and delivery behaviour

The dataset contains 50,000 records and 19 columns. It provides the attributes required to study ordering patterns and consumer preferences across multiple categories.

6. Data Preparation

- The CSV dataset was connected to Tableau as the main data source.
- Fields were reviewed and used as dimensions or measures according to their purpose.
- Required measures were aggregated for the dashboard visualizations.
- A calculated field named Age Groups was created to convert numerical age into meaningful customer segments.
- A City filter was used for dynamic city-wise analysis.
- The prepared fields were used consistently across the dashboard and Tableau Story.

Age Groups calculation:

```
IF [Age] <= 20 THEN "Child" ELSEIF [Age] <= 30 THEN "Gen Z" ELSEIF [Age] <= 40 THEN "Millennial"
ELSE "Adults" END
```

7. Tableau Visualizations

Nine main visualization areas were created in Tableau. Each visualization focuses on a different aspect of food-ordering behaviour.

Visualization	Type	Purpose
Flavors Across Cities	Heatmap	Shows cuisine distribution across cities.
Analysis of Orders Across Company Type and Meal Type	Stacked bar chart	Compares order activity by company type and meal type.
Revenue Contribution by Meal Type	Pie chart	Shows the share of revenue contributed by meal types.
Order Distribution by Age Group and Restaurant Segment	Bar chart	Compares orders across age groups and restaurant segments.
Customer Rating Analysis by Meal Type	Bar chart	Compares customer ratings across meal types.
Cities by Order Value	Treemap	Compares order value across cities.
Total Orders	Summary chart	Shows order distribution by customer age group.
Distribution of Customer Preferences across Cuisines	Bar chart	Shows the relative popularity of cuisines.
Distribution of Delivery Time Across Orders	Distribution chart	Shows the pattern of delivery time across orders.

8. Dashboard

The Tableau Dashboard combines the visualization areas into a single interactive interface. It provides a consolidated view of food-ordering behaviour, consumer preferences, ratings, revenue, city-wise demand and delivery patterns.

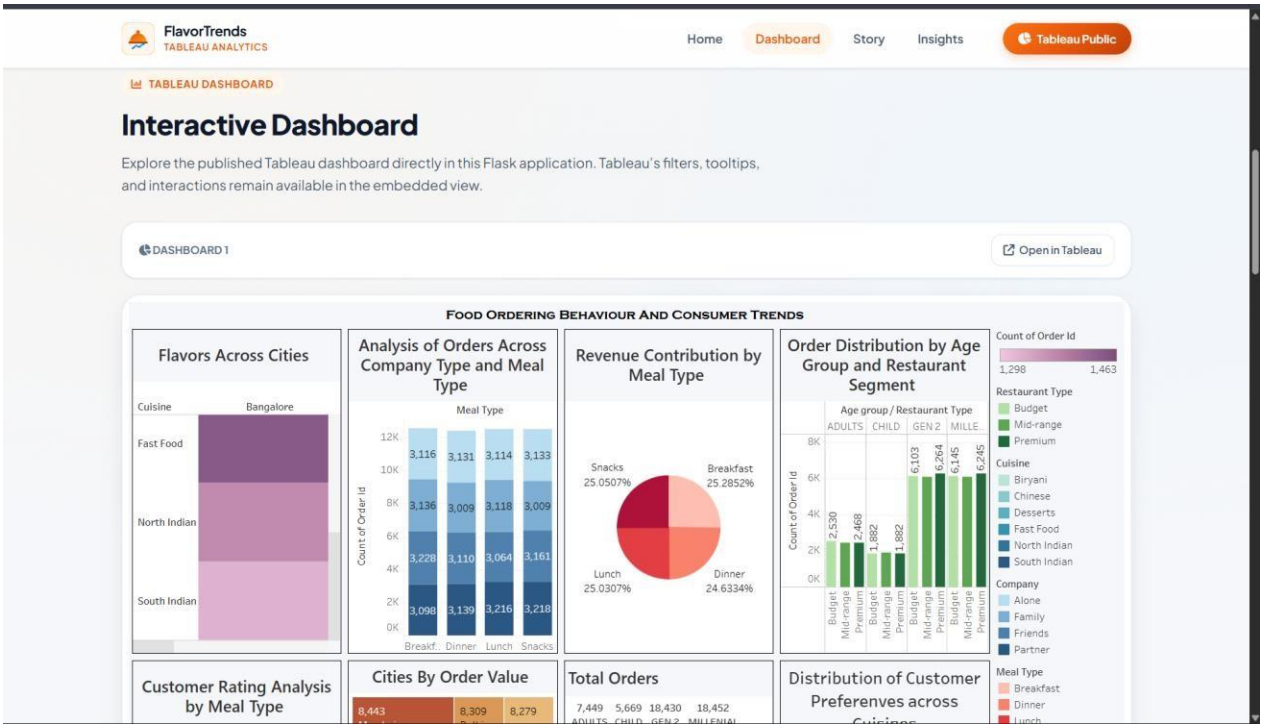


Figure 1: Food Ordering Behaviour and Consumer Trends dashboard displayed in the Flask website.

The dashboard shown above is the actual project dashboard page supplied for this documentation. It preserves Tableau interactions such as filters, tooltips and view navigation.

Key dashboard indicators shown in the project include Total Orders of 50,000, Total Revenue of ₹2,73,86,555, Average Order Value of ₹547.70 and Average Rating of approximately 2.99.

9. Tableau Story

The Tableau Story arranges the analysis into eight key insight points. Each point focuses on one important consumer or order behaviour finding.

Flavors Across Cities — Each city reveals distinct cuisine preferences that help identify local demand patterns.

Order Distribution by Age Group and Restaurant Segment — Order activity is compared across customer age groups and restaurant segments.

Analysis of Orders Across Company Type and Meal Type — The analysis compares orders across company categories and meal types.

Customer Rating Analysis by Meal Type — Average customer ratings are compared across breakfast, dinner, lunch and snacks.

Revenue Contribution by Meal Type — The analysis compares the revenue contribution of different meal types.

Distribution of Customer Preferences across Cuisines — Cuisine-wise order counts highlight relative customer preferences.

Cities by Order Value — The treemap compares order value across the analysed cities.

Distribution of Delivery Time Across Orders — Delivery-time values are analysed to understand the overall delivery pattern.

10. Flask Web Integration

A Flask-based web application was developed to integrate the published Tableau Dashboard and Story. The website provides navigation for Home, Dashboard, Story and Insights, while preserving the embedded Tableau views.

- Flask is used as the web application framework.
- The published Tableau Dashboard is embedded in the Dashboard page.
- The published Tableau Story is embedded in the Story page.
- HTML, CSS and JavaScript are used for the website structure and interaction.
- The embedded views retain Tableau's native interactive behaviour.

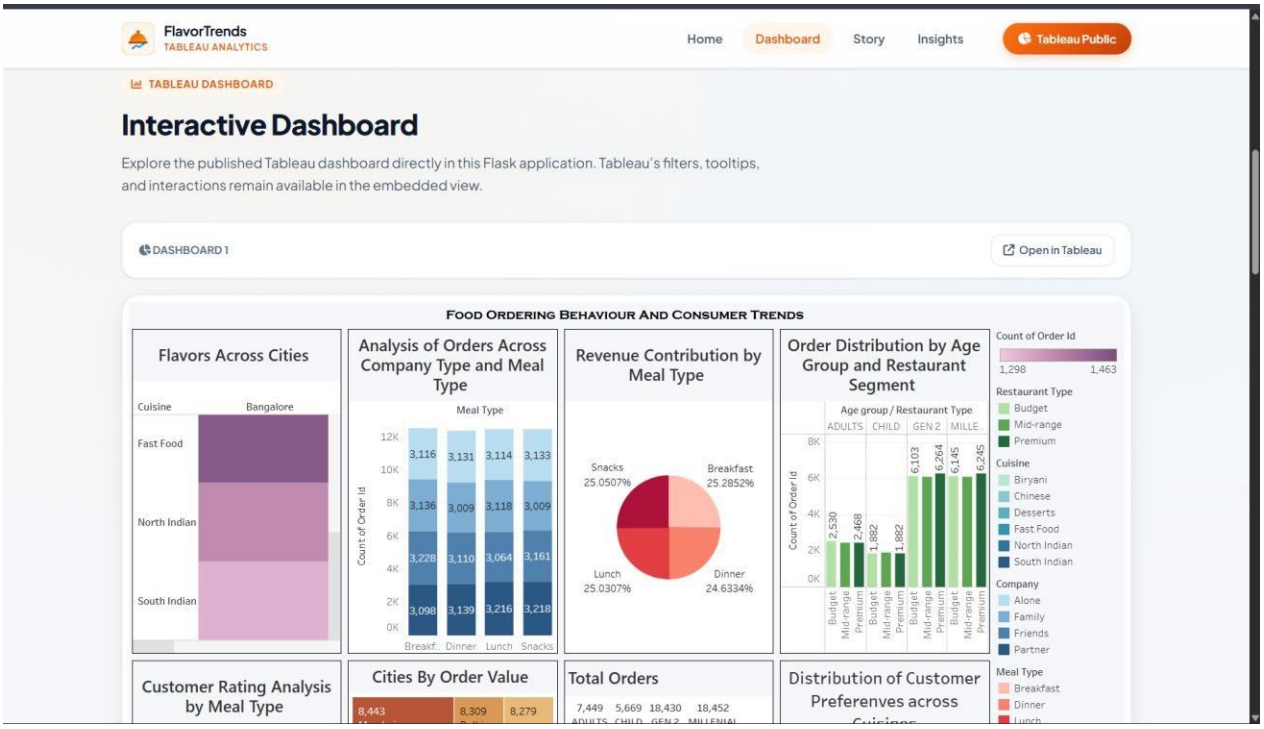


Figure 2: Dashboard view inside the Flask web application.

11. Results / Key Insights

- The dataset contains 50,000 food-order records suitable for multi-dimensional analysis.
- Cuisine preferences vary across cities, showing differences in local demand.
- Company type and meal type can be compared directly through the dashboard.
- Dinner contributes a major share of total revenue in the analysed data.
- Customer ratings show relatively close values across meal types.
- The city-wise order-value visualization highlights differences among the analysed cities.
- Cuisine preference counts show demand distributed across multiple cuisine categories.
- The delivery-time visualization provides an overall view of order delivery behaviour.

12. Performance Testing

- Amount of Data Rendered: 50,000 records across 19 columns were used for the analysis.
- Utilization of Data Filters: the City filter supports dynamic analysis of city-specific results.
- Number of Calculation Fields: the Age Groups calculated field supports age-based segmentation.
- Number of Visualizations / Graphs: nine main visualization areas are used in the dashboard.
- Interactive testing: dashboard navigation, tooltips and embedded Tableau views were checked in the Flask website.

13. Limitations

- The analysis is based on the available project dataset and therefore depends on its data quality and coverage.
- The dashboard is intended for visualization and analytical demonstration rather than a complete production food-ordering system.
- The project focuses on descriptive analysis; predictive modelling is outside the current scope.
- The Flask integration depends on the availability of the published Tableau Public views.

14. Conclusion

This project presents a complete data visualization workflow for Food Ordering Behaviour and Consumer Trends. The dataset was prepared and analysed in Tableau, transformed into multiple visualizations, combined into an interactive dashboard, and organized into an eight-point Tableau Story.

The analysis provides a clear visual understanding of cuisine preferences, meal-wise ordering and revenue patterns, age-group behaviour, customer ratings, city-wise order value and delivery-time distribution. The published Tableau work was also integrated into a Flask web application, completing the workflow from dataset to interactive web presentation.

15. Future Scope and References

Future Scope

- Connect the project to a live or regularly updated database.
- Add advanced customer segmentation and predictive analysis.
- Introduce additional filters for restaurant type, company, cuisine and time.
- Add forecasting for future order volumes and revenue.
- Deploy the Flask application publicly for wider access.
- Add downloadable analytical reports and additional business KPIs.

References

- Tableau Public – publishing and hosting of the project dashboard and story.
- Tableau Desktop / Tableau Public documentation – visualization and dashboard development.
- Flask documentation – web application framework and routing.
- HTML, CSS and JavaScript documentation – web interface development.
- Dataset: food_ordering_behaviour_dataset.csv.